



ConnectCall

Enterprise-Grade Audio/Video Calling & Screen Sharing App

GITHUB REPOSITORY LINK

<https://github.com/DeepakMewada20/connect-call.git>



Project Overview

ConnectCall provides a WhatsApp/FaceTime-grade calling experience on mobile devices. Built with **Flutter**, **Firebase**, and **ZEGOCLOUD**, it solves complex mobile real-time communication challenges including:

- Receiving and answering incoming calls when the app is in the background or killed (terminated state).
- Seamless screen sharing on Android 14+ using strict MediaProjection foreground services.
- Resilient Unicode handling preventing UTF-16 crashes when rendering contact names with emojis.
- Full offline-first local SQLite database for instant call logs without network roundtrips.



Key Features

1. Authentication & Onboarding

- **Phone Number OTP Login:** Seamless authentication with Firebase Phone Auth, automated verification, and SMS OTP resend timers.
- **User Profile Management:** Custom profile avatar upload with interactive circular image cropping (`image_cropper`).
- **Real-time Online Presence:** Automatic active/idle presence tracking stored in Cloud Firestore.

2. High-Definition Audio Calling

- **Crystal Clear Audio:** Hardware acoustic echo cancellation (AEC), automatic gain control (AGC), and active noise suppression.
- **In-Call Audio Controls:** Real-time microphone mute/unmute, audio routing toggle (Earpiece vs. Speakerphone), and dynamic soundwave animations.
- **Live Call Timer:** Precise duration tracking synchronized with local storage.

3. High-Definition Video Calling

- **1-on-1 & Multi-Party Video:** Smooth 720p/1080p adaptive bitrate video streams.
- **Camera Controls:** Instant front/rear camera flip, local video mute, and aspect-fill video rendering.
- **Gallery Layout with Fullscreen Support:** Dynamically adapts stream tiles to optimize screen space and host active screen sharing.

4. Real-time Screen Sharing

- **Android 14+ Ready:** Fully configured `mediaProjection` foreground service compliant with Android 14 security rules.
- **Interactive Toggles:** In-call quick action buttons in top and bottom bars.
- **Active Visual Indicator:** Top floating pulsing badge ( "Sharing your screen") with a single-tap "Stop" action.

5. Multi-Party Conferencing & In-Call Invitations

- **Add to Call:** Bottom sheet participant picker allowing users to invite active contacts directly into an ongoing call without leaving the room.

6. Intelligent Contact Management

- **Two-Way Sync:** Fetches device contacts with non-blocking caching.
- **Address Book Priority Resolution:** Automatically prioritizes names saved in local contacts over cloud usernames.
- **Safe UTF-16 & Emoji Support:** Custom Unicode grapheme parser preventing unpaired surrogate exceptions.
- **Smart Search & Quick Actions:** Direct integration to launch phone dialer, SMS app, or WhatsApp chat.

7. 📢 Push Notifications & Smart Call Handling

- **FCM High-Priority Data Messages:** Wakes up terminated apps on incoming calls.
- **Unified Decision Dialog:** Custom full-screen Accept / Reject dialog with a 60-second expiration guard.
- **Persistent Ringtone & Vibration:** High-priority notification channels configured with full-screen intent permissions.

8. 📄 Offline-First SQLite Call History

- **Zero Latency:** Reads and writes call logs directly into a local SQLite database (`sqlite`).
- **Comprehensive Metadata:** Tracks caller, callee, call direction, status, timestamp, and duration.
- **One-Tap Redial:** Instantly place an audio or video call directly from any history card.

🔧 SDK & Technology Stack

COMPONENT	SPECIFICATION
Framework	Flutter (Target: <code>>=3.29.0</code>)
Language	Dart (<code>>=3.12.2 <4.0.0</code>)
State Management	GetX (Reactive State, Dependency Injection & Navigation)
Target Platforms	Android (API 24 to 35 / Android 14+), iOS
Local Database	SQLite (<code>sqlite</code>)
Cloud Services	Google Firebase (Auth, Firestore, Messaging, Functions)
RTC Media Engine	ZEGOCLOUD Express Engine & ZIM Signaling

📦 Packages & Dependencies Breakdown

Here is the structured breakdown of all production and development dependencies used in `pubspec.yaml` :

1. State Management & Routing

- `get`: [^4.7.3](#) — Reactive state management, dependency injection & route navigation

2. Firebase Ecosystem

- `firebase_core`: [^4.14.0](#) — Core Firebase initialization
- `firebase_auth`: [^6.6.1](#) — Phone OTP authentication service
- `cloud_firestore`: [^6.9.0](#) — Cloud database for users & status
- `cloud_functions`: [^6.4.0](#) — Secure token server communication
- `firebase_messaging`: [^16.4.1](#) — FCM push notifications & background wake

3. Calling & RTC Media (ZEGOCLOUD)

- `zego_uikit_prebuilt_call`: [^4.24.4](#) — Prebuilt video/audio call UI
- `zego_uikit_signaling_plugin`: [^2.8.21](#) — Call invitation signaling engine
- `zego_uikit`: [^2.29.2](#) — Core RTC UI components & stream handlers

4. Local Storage & Database

- `sqflite`: [^2.4.3](#) — SQLite database for persistent offline call history
- `path`: [^1.9.1](#) — File and database path utilities
- `shared_preferences`: [^2.3.5](#) — Key-value pair local storage

5. Contacts & Permissions

- `flutter_contacts`: [^2.3.1](#) — Native contact fetching & synchronization
- `permission_handler`: [^12.0.3](#) — Runtime permissions manager

6. Media, UI & Utilities

- `image_picker`: [^1.1.2](#) — Profile avatar selection from gallery/camera
- `image_cropper`: [^12.2.1](#) — Circular image cropping widget
- `pininput`: [^6.0.2](#) — Optimized OTP input fields
- `flutter_local_notifications`: [^22.0.1](#) — Local call alerts & channels
- `url_launcher`: [^6.3.2](#) — External app launcher (Dialer, WhatsApp)
- `share_plus`: [^12.0.2](#) — Platform share sheets

7. Development Dependencies (dev_dependencies)

- flutter_test: [SDK Flutter](#) — Automated unit & widget testing SDK
- sqflite_common_ffi: [^2.4.2+1](#) — FFI implementation for desktop/unit testing SQLite
- flutter_lints: [^6.0.0](#) — Recommended Dart static code analysis rules



Architecture & Design Pattern

The application strictly follows a **Layered Clean Architecture** combined with GetX for dependency injection and reactive state bindings:

```
lib/
├── core/                          # Enterprise Foundation Layer
│   ├── constants/                # App strings, assets, dimensions
│   ├── theme/                    # Color palettes, dark/light themes
│   └── utils/                     # UTF-16 StringUtils, PhoneNumberUtil
├── data/                          # Local & Remote Persistence Layer
│   ├── datasources/              # SQLite Local Data Source (sqflite)
│   └── repositories/             # Call History Repository implementation
├── models/                       # Type-Safe Domain Data Models
├── routes/                       # Central Declarative Navigation
├── screens/                      # Presentation Layer (UI & Controllers)
└── services/                    # Singleton Business Logic & Platform APIs
```



Setup & Installation Guide

Step 1: Clone the Repository

```
git clone https://github.com/DeepakMewada20/connect-call.git
cd connect_call
```

Step 2: Install Dependencies

```
flutter pub get
```

Step 3: Configure Firebase & ZEGOCLOUD

- Create a project on the Firebase Console and enable Phone Authentication.
- Place your Android `google-services.json` inside `android/app/google-services.json`.
- Deploy token server Cloud Functions for ZEGOCLOUD security tokens.

Step 4: Build & Install

```
flutter run
# Or build release APK
flutter build apk --release
```



Testing Matrix

Automated Test Suite Status: 201 Passing Tests (100% Core Coverage)

```
flutter test
# Expected Output:
00:14 +201: All tests passed!
```



AI Engineering Tools

This application was architected and optimized using **Google DeepMind Advanced Agentic Coding (Antigravity AI)** for autonomous bug diagnosis, TDD test suite generation (201 tests), UTF-16 Unicode surrogate sanitization, and Android 14 MediaProjection compliance.